

CLASS X PRACTICE QUESTION PAPER

CBSE-STYLE PRACTICE EXAMINATION

SUBJECT: MATHEMATICS (STANDARD)

Time Allowed: 1.5 Hours

Maximum Marks: 40

General Instructions:

1. This question paper contains three sections: Section A, Section B, and Section C.
2. Section A consists of Multiple Choice Questions (MCQs).
3. Section B consists of Short Answer Type Questions.
4. Section C consists of Long Answer Type Questions.
5. All questions are compulsory. Marks for each question are indicated on the right.

SECTION A - MULTIPLE CHOICE QUESTIONS

(14 Questions * 15 Marks)

- Q1.** The area of a quadrant of a circle whose circumference is 22 cm is (use $\pi = 22/7$): [1]
(A) $77/8 \text{ cm}^2$
(B) $77/4 \text{ cm}^2$
(C) $77/2 \text{ cm}^2$
(D) 154 cm^2
- Q2.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes (use $\pi = 22/7$). [2]
(A) $154/3 \text{ cm}^2$
(B) $154/6 \text{ cm}^2$
(C) $308/3 \text{ cm}^2$
(D) $77/3 \text{ cm}^2$
- Q3.** A line intersecting a circle in two points is called a: [1]
(A) Tangent
(B) Secant
(C) Chord
(D) Diameter
- Q4.** If TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to: [1]
(A) 60°
(B) 70°
(C) 80°
(D) 90°
- Q5.** The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has: [1]
(A) A unique solution
(B) Exactly two solutions
(C) Infinitely many solutions
(D) No solution
- Q6.** If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is: [1]
(A) $-5/4$
(B) $2/5$
(C) $15/4$
(D) $3/2$
- Q7.** Which of the following quadratic equations has no real roots? [1]
(A) $2x^2 - 3x + 1 = 0$
(B) $x^2 - 4x + 4 = 0$
(C) $x^2 + 2x + 5 = 0$
(D) $2x^2 - 6x + 3 = 0$

- Q8.** The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is: [1]
 (A) -8
 (B) 10
 (C) -16
 (D) 8
- Q9.** The angle of elevation of the top of a tower from a point on the ground, 30 m away from the foot of the tower, is 30 deg. [1]
 The height of the tower is:
 (A) $10\sqrt{3}$ m
 (B) $30\sqrt{3}$ m
 (C) $20\sqrt{3}$ m
 (D) 10 m
- Q10.** The angle of elevation of the top of a 30 m high tower from a point on the ground is 30 deg. The distance of the point [1]
 from the base of the tower is:
 (A) 30 m
 (B) $30\sqrt{3}$ m
 (C) $10\sqrt{3}$ m
 (D) 60 m
- Q11.** The empirical relationship between the three measures of central tendency is: [1]
 (A) $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$
 (B) $\text{Mode} = 3 \text{ Mean} - 2 \text{ Median}$
 (C) $2 \text{ Median} = 3 \text{ Mode} - \text{Mean}$
 (D) $\text{Mean} = 3 \text{ Median} - 2 \text{ Mode}$
- Q12.** The modal class for the following frequency distribution: Classes: 0-10 ($f=5$), 10-20 ($f=8$), 20-30 ($f=15$), 30-40 ($f=10$), [1]
 40-50 ($f=6$) is:
 (A) 20
 (B) 30
 (C) 40
 (D) 50
- Q13.** 2 cubes each of volume 64 cm^3 are joined end to end. The surface area of the resulting cuboid is: [1]
 (A) 128 cm^2
 (B) 160 cm^2
 (C) 192 cm^2
 (D) 256 cm^2
- Q14.** A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. The height of the [1]
 cylinder is:
 (A) 74 cm
 (B) 24 cm
 (C) 14 cm
 (D) 84 cm

SECTION B - SHORT ANSWER QUESTIONS

(5 Questions * 15 Marks)

- Q15.** If the sum of the first 14 terms of an AP is 1050 and its first term is 10, find the 20th term. [3]
- Q16.** If the squared difference of the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ is equal to 144, find the value of [3]
 p.
- Q17.** A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the [3]
 probability that it bears: (i) a two-digit number, (ii) a perfect square number, (iii) a number divisible by 5.
- Q18.** Prove that $3 + 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. [3]
- Q19.** In ΔABC , D is a point on side BC such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \times CD$. [3]

SECTION C - LONG ANSWER QUESTIONS

(2 Questions * 10 Marks)

Q20. If A(-5, 7), B(-4, -5), C(-1, -6) and D(4, 5) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD. [5]

Q21. If $\sec \theta + \tan \theta = p$, obtain the values of $\sec \theta$, $\tan \theta$, and $\sin \theta$ in terms of p . [5]

*** END OF QUESTION PAPER ***